

# Working Questions – Week 5 (Internet Programming)

**Topics:** Responsive design, `<meta name="viewport">`, relative font sizes (`em`), background properties, `calc()`, and CSS positioning (`static/fixed/absolute/relative`), Flexbox basics.

---

## Part A – Multiple Choice (5 Questions)

- Q1.** To prevent mobile browsers from rendering the page at desktop width and zooming out, which tag should be placed inside `<head>`?
- a) `<meta charset="UTF-8">`
  - b) `<meta name="viewport" content="width=device-width, initial-scale=1">`
  - c) `<meta http-equiv="refresh" content="30">`
  - d) `<meta name="robots" content="noindex">`
- Q2.** In CSS, `1em` is usually:
- a) Always 16px
  - b) Equal to the element's current font size (relative)
  - c) Equal to the screen width
  - d) Equal to the browser zoom level only
- Q3.** Which background setting makes an image fill the element while keeping aspect ratio, possibly cropping the image?
- a) `background-size: contain;`
  - b) `background-size: cover;`
  - c) `background-repeat: no-repeat;`
  - d) `background-position: center;`

- Q4.** Which CSS function lets you define a value using an expression like `width: calc(50% - 10px);` ?
- a) `var()`
  - b) `calc()`
  - c) `minmax()`
  - d) `url()`
- Q5.** Which `position` value offsets an element **from its normal location** without removing it from the document flow?
- a) `static`
  - b) `fixed`
  - c) `absolute`
  - d) `relative`

## Part B – Fill in the Blanks (5 Questions)

- B1.** The viewport tag should be placed inside the \_\_\_\_\_ section of an HTML file.
- B2.** The unit `em` is a \_\_\_\_\_ (absolute / relative) unit.
- B3.** To show a background image **once** and prevent tiling, use: \_\_\_\_\_.
- B4.** To place a background image at the top of an element, use: \_\_\_\_\_.
- B5.** In positioning, `top`, `left`, `right`, `bottom` offsets work only when `position` is \_\_\_\_\_ (not `static`).
- 

## Part C – Coding (Adjust the Given Code) (4 Exercises)

**Instructions:** For each exercise, you are given HTML/CSS. Do **only** the requested changes.

### C1) Form – Add Responsive Viewport + Make Inputs Full Width

**Task:**

- Add the correct `viewport` meta tag into the `<head>`.
- Make the text inputs and button fill the available width.
- Use `em` for padding (example: `padding: 0.8em;`).

**Given Code:**

```

<!-- index.html -->
<!DOCTYPE html>
<html>
<head>
  <title>Login</title>
</head>
<body>
  <div class="card">
    <h2>Login</h2>
    <form>
      <label>Email</label>
      <input type="email" placeholder="name@example.com">

      <label>Password</label>
      <input type="password" placeholder=".....">

      <button type="submit">Sign In</button>
    </form>
  </div>
</body>
</html>

/* styles.css */
.card{
  width: 380px;
  padding: 20px;
  border: 1px solid #ccc;
}
input{
  padding: 10px;
}
button{
  padding: 10px;
}

```

**Expected adjustments (hint):** add `width: 100%;` and `box-sizing: border-box;` to `inputs/button`.

## C2) Form – Use `calc()` for Two Inputs in One Row + Position Icon

### Task:

- Put `First name` and `Last name` inputs on the same row.
- Use `calc()` so each input is half width with a small gap (example: `width: calc(50% - 8px);`).
- Add a small icon inside the input area using `position: absolute;` (icon should appear on the right side).

## Given Code:

```
<!-- signup.html -->
<div class="card">
  <h2>Sign Up</h2>

  <div class="row">
    <div class="field">
      <label>First name</label>
      <input type="text" placeholder="Ada">
      <span class="icon"></span>
    </div>

    <div class="field">
      <label>Last name</label>
      <input type="text" placeholder="Lovelace">
      <span class="icon"></span>
    </div>
  </div>

  <button>Create Account</button>
</div>

/* signup.css */
.card{
  width: 520px;
  border: 1px solid #ddd;
  padding: 16px;
}
.row{
  /* currently not aligned */
}
.field{
  margin-bottom: 12px;
}
.icon{
  /* currently appears under the input */
}
```

**Hint:** Make `.row` a flex container OR use inline-block. Also set `.field` to `position: relative`; and `.icon` to `position: absolute; right: ...; top: ...;`.

## C3) Flexbox – Center Items and Add Space Between

### Task:

- Use Flexbox to place the logo on the left and the links on the right.
- Vertically center all items.

- Add equal spacing between links.

### Given Code:

```
<!-- nav.html -->
<header class="nav">
  <div class="logo">MySite</div>
  <nav class="links">
    <a href="#">Home</a>
    <a href="#">Projects</a>
    <a href="#">Contact</a>
  </nav>
</header>
```

```
/* nav.css */
.nav{
  border-bottom: 1px solid #ccc;
  padding: 12px;
}
.links a{
  text-decoration: none;
}
```

**Hint:** Use `display: flex; justify-content: space-between; align-items: center;` and `gap: ...;` for the links.

## C4) Flexbox – Make a Responsive Card Grid + Use em Fonts

### Task:

- Display cards in a row that wraps on small screens (responsive).
- Each card should have a flexible width (example: `flex: 1 1 220px;`).
- Use `em` to increase the card title size (example: `font-size: 1.2em;`).

## Given Code:

```
<!-- cards.html -->
<section class="grid">
  <article class="card"><h3>Card 1</h3><p>Text...</p></article>
  <article class="card"><h3>Card 2</h3><p>Text...</p></article>
  <article class="card"><h3>Card 3</h3><p>Text...</p></article>
  <article class="card"><h3>Card 4</h3><p>Text...</p></article>
</section>
```

```
/* cards.css */
.grid{
  border: 1px dashed #bbb;
  padding: 12px;
}
.card{
  border: 1px solid #ddd;
  padding: 12px;
}
```

**Hint:** Use `display:flex`; `flex-wrap:wrap`; `gap:...`; on `.grid`.

---

## End of Week 5 Working Questions